

Multi-Representational Memory Discovery for Progressive Retrieval Attention

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Abstract

Progressive Retrieval Attention (PRA) exposes a large logical memory while admitting only a small native-key/value working set to attention. That design moves a basic question upstream: how should the system discover the few memory objects worth materializing? We compare contextual semantic gists with exact model-token identity, BM25, approximate token matching, and state-dependent hybrid search. The representations fail differently. In a fresh Gate B evaluation with 20 validation and 50 held-out identities per dataset, Qwen3-0.6B and SmolLM2-135M reproduce the same dataset-level ordering: BM25 leads HotpotQA evidence recall at 0.665 and 0.662, whereas approximate matching leads QASPER at 0.747 and 0.783. A sparse posting-list prototype preserves exhaustive Top-4 output for every paired query while scoring about one tenth of candidates and reducing mean Python routing time by 68.4x; at 4,096 chunks, exact indexed lookup is 283–300x faster. Fuzzy paths remain too slow for production serving. We then carry selected identities through a matched four-chunk, 128-token native-K/V materializer. On HotpotQA, correct memory improves frozen-model gold-answer likelihood, and an evidence oracle beats irrelevant memory for both model families. On QASPER, retrieval transfers across models but fixed-channel likelihood gains over disabled memory remain unresolved, even though oracle memory still beats irrelevant memory. Generated EM/F1 is weak throughout. A broader four-dataset study further shows that root and successor search prefer different representations on three datasets. The resulting conclusion is deliberately split: multi-representational sparse discovery is feasible, but retrieval success and useful model consumption are separate problems. PRA should expose retrieval representation as an explicit action rather than hide it inside one universal score.

1 Introduction

An addressable memory is useful only if the system can find the right address. A semantic retriever can recognize that two passages discuss related ideas, but relatedness is not identity. The string CVE-2026-1234, a DOI, a URI, or a filename may identify one object exactly. An alias such as “IBM” may identify the same organization as “International Business Machines.” A question such as “Which university did the founder of company X attend?” instead asks for a relation whose answer need not repeat the question’s words. These cases motivate the boundary

$$\boxed{\text{semantic relatedness} \neq \text{referential identity}}.$$

PRA treats long context as logical memory rather than requiring every source token to remain active at every decoding step [11, 12]. A compact representation first discovers candidate memory objects; a materializer then admits bounded native K/V; finally, the frozen model must use that

state. This paper studies the first step and tests its handoff to the other two. It does not assume that higher retrieval recall implies better answers.

Consider a two-hop query with an evidence path

$$Q \rightarrow C_1 \rightarrow C_2.$$

The root C_1 might be semantically related to the question. Once admitted, however, it may expose the exact name or identifier of C_2 . The useful search representation can therefore change with state:

$$S_{\text{root}} \neq S_{\text{succ}}.$$

The same distinction separates this work from static sparse–dense fusion. We do not propose hybrid information retrieval as a novelty. We ask which retrieval representation should address identity-aligned PRA memory at each stage, whether those choices can be indexed sparsely, and whether the selected memory produces a causal effect after native-K/V admission.

The paper answers four questions. First, no single measured representation suffices: semantic, exact, BM25, and approximate channels recover different evidence. Second, this heterogeneity replicates on fresh identities across two tested model/tokenizer families. Third, posting-list candidate generation preserves Top-4 retrieval in the measured suite while touching about 10% of candidates on average. Fourth, correctly discovered HotpotQA memory produces useful frozen-model likelihood effects, whereas QASPER demonstrates that strong retrieval can coexist with unresolved fixed-channel consumption.

Our contributions are:

1. a multi-representational PRA discovery formulation that separates semantic relatedness from token and reference identity;
2. fresh cross-model evidence that retrieval representation is heterogeneous, with BM25 leading HotpotQA and approximate matching leading QASPER;
3. sparse indexed candidate generation that preserves measured Top-4 behavior while substantially reducing prototype routing work;
4. a matched native-K/V Gate B showing causal HotpotQA likelihood gains and a QASPER retrieval-to-consumption counterexample;
5. root/successor and interaction evidence showing dependence on stage, budget, granularity, and model; and
6. a versioned retrieval-action and provenance interface for downstream adaptive control.

2 PRA Memory Discovery

PRA separates three questions that ordinary retrieval pipelines often conflate:

1. *Discovery*: which logical memory object is relevant?
2. *Materialization*: which native K/V, at which layers and detail, enters the bounded working set?
3. *Consumption*: does the model use that admitted state beneficially?

Figure 1 shows the experimental boundary. Logical sources are chunked and indexed once. A query chooses a retrieval representation, sparse candidate generation narrows the logical address space, and exact scoring returns stable identities. The inherited materializer resolves those identities to native K/V under a physical token budget. The frozen model then supplies a separate consumption measurement.

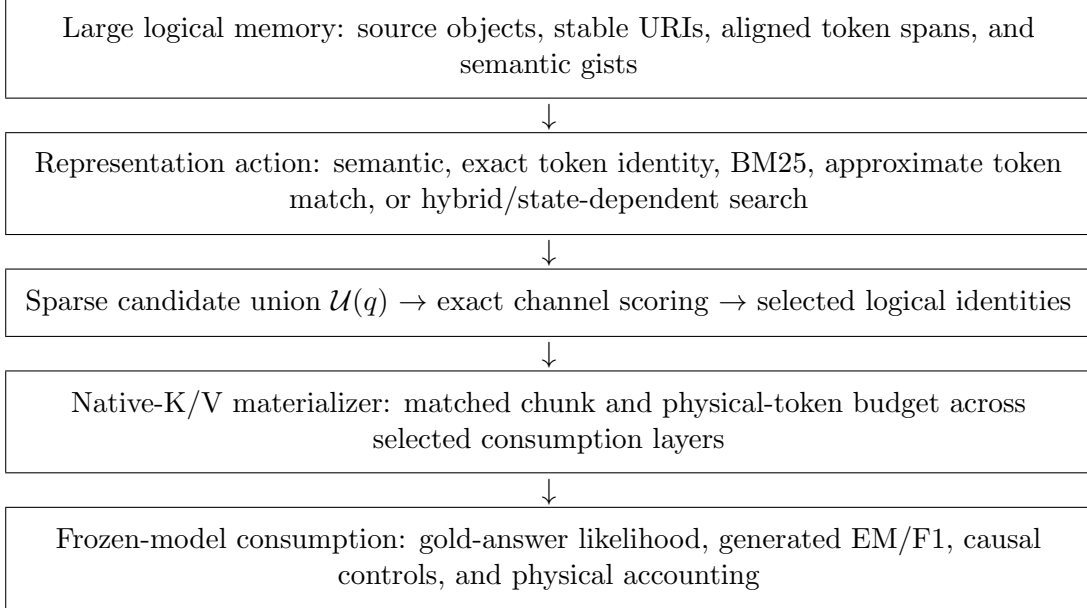


Figure 1: Discovery, materialization, and consumption are separate interfaces. Paper 2.6 changes the representation used to discover logical identities, holds native-K/V materialization fixed in Gate B, and then measures whether the frozen consumer benefits.

For evaluation, let E be the required evidence identities and K the deduplicated selected set. We introduce three quality quantities together:

$$R_E = \frac{|K \cap E|}{|E|}, \quad P_E = \frac{|K \cap E|}{|K|}, \quad C_E = \mathbf{1}[E \subseteq K].$$

Candidate-normalized cost is reported separately as K/N , where N is the logical candidate count, and K/E , the selected working-set overhead relative to required evidence. We never substitute candidates scored for chunks selected, physical K/V tokens, layer-token states, or wall-clock time.

3 Retrieval Representations

Table 1 summarizes the primary actions. All channels return stable row identities and retain provenance. The semantic channel scores a contextual query $q_{t,\ell}$ against layer-aligned chunk gist $g_{i,\ell}$,

$$s_{i,\ell}^{\text{sem}} = \cos(q_{t,\ell}, g_{i,\ell}).$$

Exact token spans preserve literal identity. BM25 rewards informative decoded terms. Approximate matching uses character-trigram candidate narrowing before bounded sequence comparison. Hybrid search can combine channels or choose different representations at root and successor stages. Explicit registered URIs form a deterministic product path when a caller already possesses the address; they are not a learned or oracle result.

Table 1: Primary retrieval representations and their characteristic strengths and failures.

Representation	Useful signal	Characteristic failure
Semantic gist	contextual topic and implicit relation	literal identifiers can be blurred
Exact token	literal span, URI, filename, numeric ID	paraphrase, typo, and changed segmentation
BM25	several moderately informative lexical terms	vocabulary mismatch and implicit relation
Approximate token	typo, partial mention, surface variation	collisions, stale aliases, wrong instance
Hybrid/stateful	complementary channels and newly exposed state	distractor dilution and score calibration

The token-native sidecar uses the host tokenizer rather than a second NLP vocabulary. It stores raw token IDs, normalized pieces, token n -grams, BM25 terms, character trigrams, stable URIs, and row-aligned source offsets. Unicode NFKC and case folding repair compatibility forms, while aliases can come from headings, parenthetical names, or an application registry. Bounded token edit distance and mean host-input-embedding similarity are diagnostic extensions rather than silent defaults.

3.1 Sparse Indexed Candidate Generation

Exhaustive Python scoring makes routing cost proportional to the full logical memory even when only a few rows share query evidence. The indexed path instead constructs a deduplicated candidate union

$$q \rightarrow \mathcal{U}(q) = \mathcal{P}_{tok} \cup \mathcal{P}_{ngram} \cup \mathcal{P}_{bm25} \cup \mathcal{P}_{char} \cup \mathcal{P}_{uri}, \quad |\mathcal{U}(q)| \ll N,$$

then applies the requested channel only inside the bounded pool. A reserved semantic slice prevents lexical postings from removing every semantically plausible row. Stable identities deduplicate cross-channel unions before ranking.

Across 256, 1,024, and 4,096 chunks and both tokenizers, a 64-row pool reproduces exhaustive Top-4 output for every paired query. Averaged over the scaling suite, the indexed path scores 9.99% of candidates and is 68.36x faster than the matching exhaustive Python implementation. At 4,096 chunks, exact lookup falls from 33.6–37.0 s to 119–125 ms, a 283–300x speedup. Weighted, approximate, and cascade paths improve by 56–64x but still take 0.50–0.65 s per query. These are mechanism and prototype cost results, not production-serving claims.

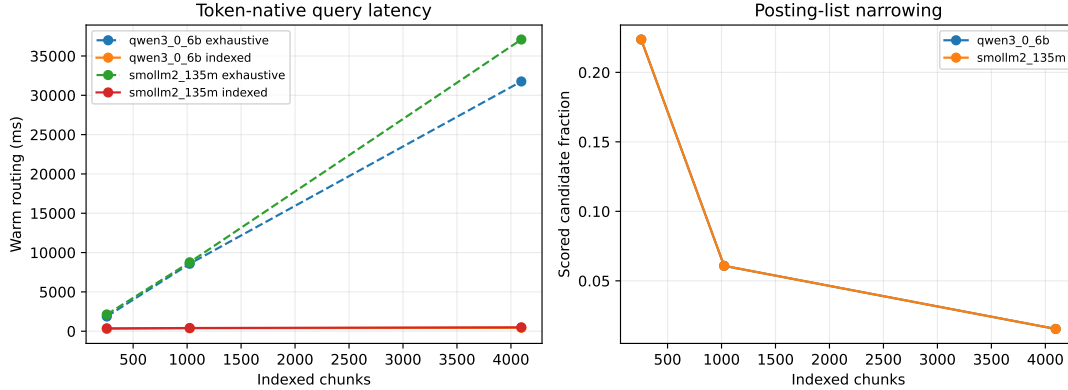


Figure 2: Exhaustive and indexed routing latency. Posting-list candidate narrowing keeps the scored pool bounded as logical memory grows. Exact lookup scales well in this prototype; fuzzy paths remain too slow for serving claims.

The immutable index rebuilds corpus statistics when references are added. At 4,096 chunks, cold construction takes 3.37 s for Qwen and 2.76 s for SmolLM; adding 64 chunks by full-statistics rebuild takes about 0.80 s. Serialized payload size is 10.06 and 10.94 MiB, excluding Python-object overhead. Appendix G gives the detailed timing and implementation contract.

4 Main Evaluation: Fresh Two-Model Gate B

4.1 Protocol

Gate B is the primary HotpotQA/QASPER evaluation. It uses Qwen3-0.6B and SmolLM2-135M with identities disjoint from the earlier discovery cohort. For each dataset, 20 examples tune only observable selector parameters and 50 fresh examples form the held-out test. A serialized audit confirms no overlap among the 48 legacy, 40 validation, 100 test, and 16 interaction identities.

The mechanism construction places annotated evidence strings first, appends the original source, bounds the result to 1,536 tokens, and partitions it into 32-token chunks. Gold labels do not enter fixed or learned channel scores, but evidence-first construction is annotation dependent; Gate B is not an unmodified QA benchmark. Every routed condition receives four chunks, at most 128 unique native-K/V tokens, and the same four late consumption layers. Disabled and empty controls use no memory; shuffled and irrelevant controls preserve the memory budget with wrong content; bounded direct context places the source in the ordinary prompt and is not compute matched. The oracle is an annotated ceiling.

4.2 Retrieval Heterogeneity Replicates

Table 2 gives the complete fixed-channel comparison. Dataset-level ordering is stable across two model/tokenizer families: BM25 leads HotpotQA at 0.665 and 0.662 recall, whereas approximate matching leads QASPER at 0.747 and 0.783. The five-seed validation-trained selector remains below those strongest fixed channels; its Qwen recall ranges only 0.540–0.560 on HotpotQA and 0.556–0.576 on QASPER. The deficit is therefore not explained by one unlucky initialization.

Table 2: Gate B held-out evidence recall, 50 examples per dataset and model. Bold marks the best fixed channel in each row. All routes select at most four chunks.

Model	Dataset	Semantic	Exact	BM25	Approx.	Hybrid	Adaptive
Qwen	HotpotQA	.473	.482	.665	.593	.505	.543
Qwen	QASPER	.364	.619	.228	.747	.457	.576
SmolLM	HotpotQA	.320	.450	.662	.593	.422	.518
SmolLM	QASPER	.348	.609	.328	.783	.437	.446

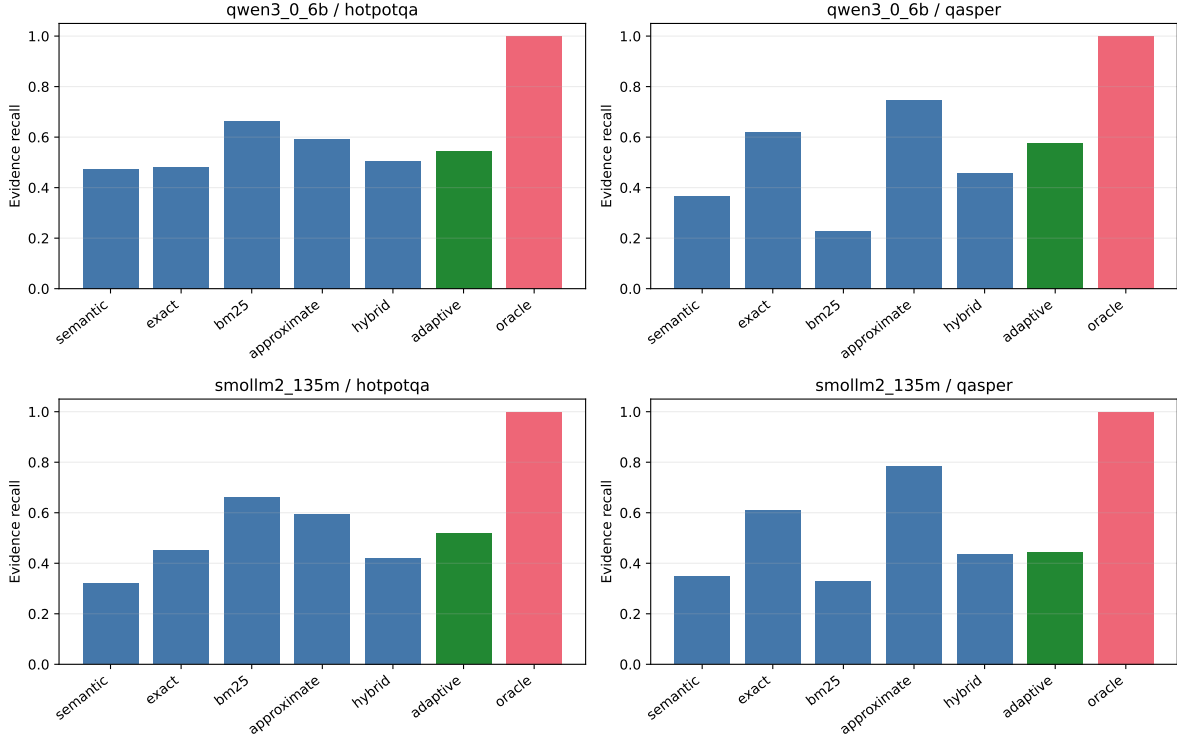


Figure 3: Cross-model Gate B evidence recall. Retrieval preference follows the dataset more consistently than the model family, while the observable selector stays below the strongest fixed channel.

4.3 Correct Memory Helps on HotpotQA

Retrieval becomes useful to PRA only when the frozen model responds beneficially after native-K/V materialization. Table 3 reports the paired likelihood effects that test this boundary. On HotpotQA, semantic memory improves mean gold-answer token log probability over disabled PRA by 0.744 for Qwen and 0.185 for SmolLM; both paired bootstrap intervals exclude zero. More stringently, oracle evidence beats equal-budget irrelevant memory by 0.770 and 0.256. Correct memory identity therefore matters after admission rather than merely perturbing logits because extra K/V exists.

Table 3: Gate B causal consumption effects. Values are paired mean changes in gold-answer token log probability with 95% bootstrap intervals.

Dataset	Model	Contrast	Effect [95% CI]
HotpotQA	Qwen	semantic memory – disabled	+.744 [+ .414, +1.132]
HotpotQA	SmolLM	semantic memory – disabled	+.185 [+ .038, +.345]
HotpotQA	Qwen	oracle evidence – irrelevant memory	+.770 [+ .508, +1.052]
HotpotQA	SmolLM	oracle evidence – irrelevant memory	+.256 [+ .163, +.354]
QASPER	Qwen	approximate memory – disabled	−.013 [−.190, +.185]
QASPER	SmolLM	approximate memory – disabled	−.008 [−.138, +.123]
QASPER	Qwen	oracle evidence – irrelevant memory	+.154 [+ .074, +.231]
QASPER	SmolLM	oracle evidence – irrelevant memory	+.091 [+ .033, +.150]

4.4 QASPER Separates Retrieval from Consumption

QASPER approximate matching recovers roughly three quarters of evidence for both models, yet its likelihood change relative to disabled memory is unresolved. The oracle still beats irrelevant memory for Qwen and SmolLM. The materializer and consumer can therefore discriminate maximally correct from wrong memory, but the strongest fixed retriever does not establish a benefit over no memory. In concise form,

$$\text{retrieval success} \not\Rightarrow \text{useful consumption}.$$

This counterexample redirects later work toward admission, attention competition, consumer adaptation, and PRA-aware training rather than treating higher recall as sufficient.

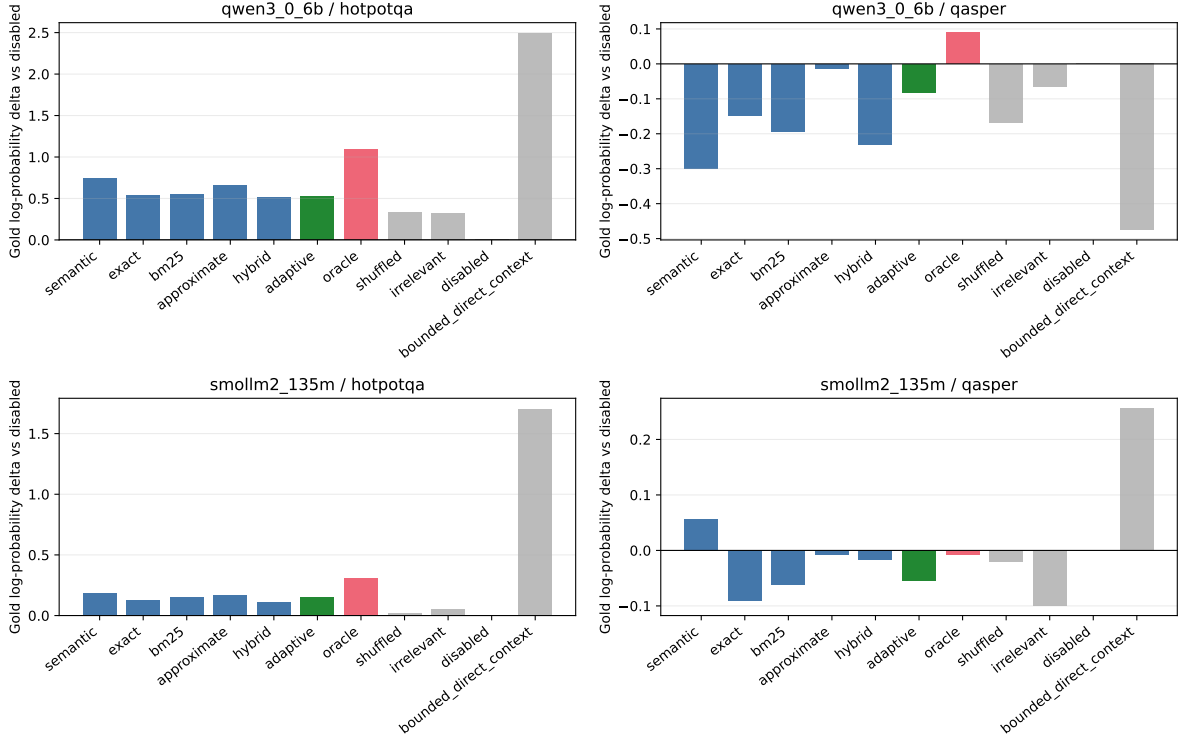


Figure 4: Gold-answer likelihood under matched discovery, oracle, wrong-memory, disabled, and bounded direct-context conditions. Direct context is a heavier comparison, not a matched-K/V condition.

Physical accounting confirms the intended match. Routed conditions materialize 127.8–128.0 unique tokens, 8.33–9.36% of the bounded source, at each of four late layers. This is approximately 512 layer-token states, or 1.05 MiB of fp16 K/V for Qwen and 0.39 MiB for SmolLM under their grouped-query head geometries. Disabled and empty conditions report no active K/V.

Generated answers remain weak. Qwen HotpotQA memory conditions obtain EM 0.00–0.02 and F1 0.074–0.120; SmolLM EM is zero. Bounded direct context reaches Qwen HotpotQA F1 0.183 but takes 5.28 s per example, versus about 1.4 s for 128-token PRA memory. These models make teacher-forced likelihood the more sensitive causal measure. We do not reinterpret it as a QA-accuracy gain.

5 Breadth: Four-Dataset Channel Geometry

Gate B establishes replication and consumption on HotpotQA/QASPER. The earlier frozen-feature cohort adds breadth across QASPER, HotpotQA, 2Wiki, and MuSiQue while holding model, layer, chunk size, and four-chunk request budget fixed. Table 4 shows a different winning channel on three task regimes: QASPER favors exact spans, HotpotQA and 2Wiki favor BM25, and MuSiQue favors approximate matching. Static and iterative hybrid search improve over semantic gists in several rows but never dominate the best individual channel.

Table 4: Legacy held-out recall/precision at four requested 32-token chunks. This breadth study is secondary to fresh Gate B.

Dataset (n)	Gist	Exact	BM25	Approx.	Static hybrid	Iterative hybrid
QASPER (16)	.038/.062	.178 /.312	.042/.094	.037/.078	.044/.078	.081/.125
HotpotQA (16)	.151/.172	.287/.328	.398 /.453	.301/.359	.288/.344	.279/.312
2Wiki (24)	.132/.188	.196/.240	.262 /.312	.203/.250	.203/.250	.186/.250
MuSiQue (18)	.108/.333	.136/.417	.132/.431	.156 /.514	.151/.500	.146/.472

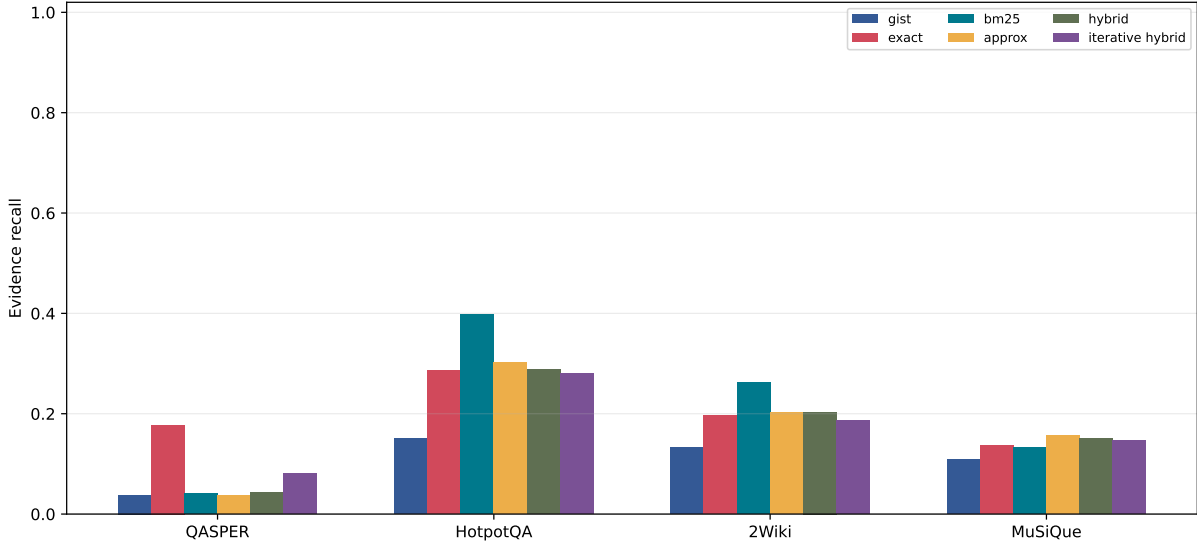


Figure 5: Matched-budget recall in the four-dataset breadth cohort. No representation dominates all cohorts, and static fusion does not reliably beat the strongest individual channel.

The representations are genuinely nonredundant. Mean selected-set Jaccard across channel pairs is 0.256 on QASPER, 0.325 on HotpotQA, 0.369 on 2Wiki, and 0.326 on MuSiQue. Yet complementarity alone does not make fusion better: at a two-chunk root budget, lexical channels add 0.188–0.812 unique gold chunks per example relative to gist search while adding 2.778–3.500 unique distractors. Reciprocal-rank fusion avoids raw-score calibration but also fails to dominate. Meaning and identity are complementary signals; their distractors are complementary too.

6 Root and Successor Search Are Different Decisions

We next divide the four-chunk budget into two root and two successor chunks. Every root representation is crossed with every successor representation, but the successor headline fixes the root policy selected on validation. For 2Wiki and MuSiQue, successor truth follows annotated dependency nodes. QASPER and HotpotQA provide unordered evidence remainders, so their successor rows diagnose staged retrieval without asserting a gold dependency edge.

Table 5: Best held-out root and successor representations under the validation-selected root. R/P are stage-local evidence recall and precision.

Dataset	Root representation	Root R/P	Successor representation	Successor R/P
QASPER	exact	.096/.312	exact new address	.025/.125
HotpotQA	exact	.047/.094	native semantic	.135/.281
2Wiki	semantic gist	.076/.125	exact new address	.050/.083
MuSiQue	approximate	.084/.139	native semantic	.032/.111

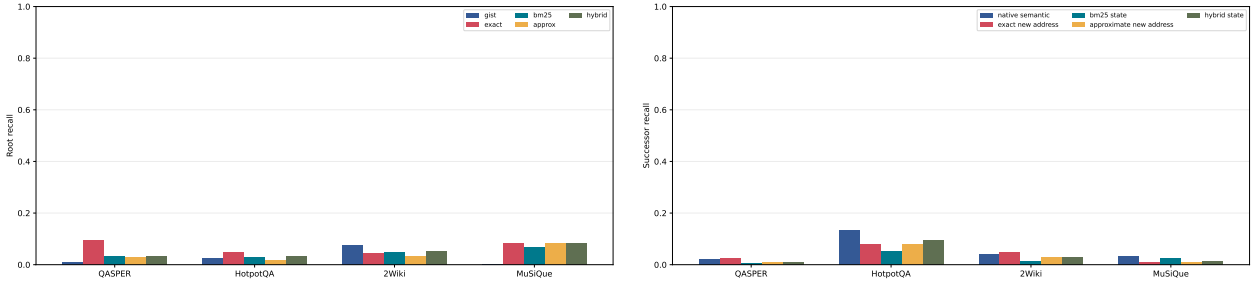


Figure 6: Root recall at two chunks and successor recall at two chunks under a fixed root policy. The preferred representation changes after admission on three of four datasets.

QASPER retains exact identity at both stages. HotpotQA and MuSiQue prefer lexical root entry but semantic successor expansion. 2Wiki prefers semantic entry and an exact newly exposed address. No per-example oracle transition occupies a majority. These low absolute recalls do not show that the staged router solves multi-hop retrieval; they show that forcing root and successor through one score discards a measurable action dimension.

Simple observable selectors do not recover the oracle. A validation-trained linear selector remains below the best held-out fixed channel on three datasets and ties only MuSiQue. Root disagreement has correlation 0.052 with true per-example oracle headroom. Paper 2.6 therefore exports S_{root} and S_{succ} as actions rather than claiming adaptive control. Full oracle, selector, confidence, and transition diagnostics appear in Appendices C and H.

7 Representation, Budget, Granularity, and Model

The interaction study crosses eight channels with 16-, 32-, and 64-token chunks, Top-2/4/8 selection, early/late routing layers, two datasets, and two model families. Its eight examples per dataset are disjoint from the legacy, validation, and Gate B sets. Because it contains only 16 identities, it diagnoses coupling rather than replacing Gate B.

At 32-token granularity, broader selection buys coverage at the cost of selectivity. Averaged over channels, late-layer Qwen HotpotQA recall rises from 0.562 to 0.703 to 0.883 as Top- K grows from 2 to 4 to 8; complete recovery rises from 0.406 to 0.562 to 0.812, while precision falls from 0.414 to 0.229. SmolLM follows the same pattern, reaching 0.953 recall and 0.906 complete recovery at Top-8 while precision falls to 0.225. QASPER reaches 0.917 and 0.943 recall at Top-8 for Qwen and SmolLM. Thus

breadth buys coverage at the cost of selectivity.

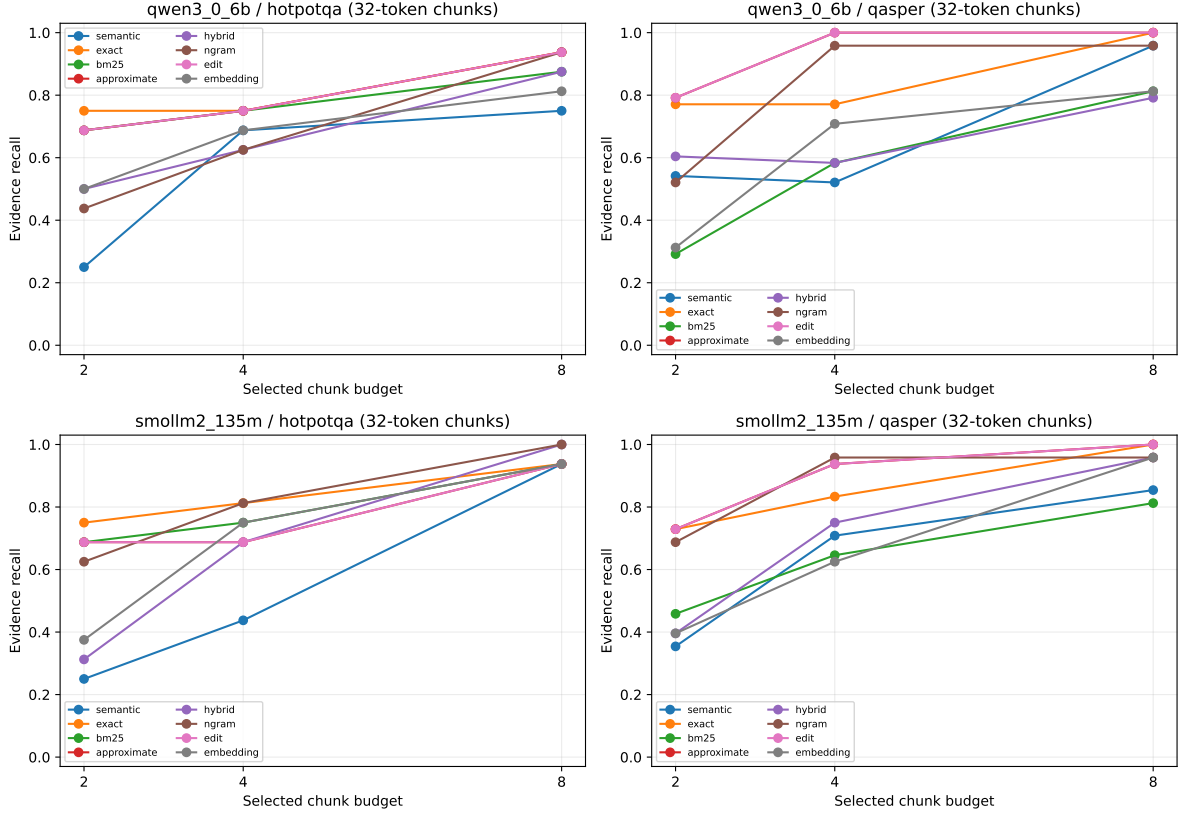


Figure 7: Evidence recall as the reachable selected-chunk budget grows at fixed 32-token granularity. Additional chunks recover more evidence and admit more distractors.

Holding routed physical tokens fixed at 128 reveals a second interaction. The best channel changes with chunk size, dataset, and model. Qwen/HotpotQA favors token n -grams at 16 and 64 tokens but bounded edit at 32; SmolLM/QASPER favors hybrid at 16, token n -grams at 32, and exact at 64. Token-native channels are layer invariant by construction, but semantic recall changes between measured early and late layers by $+0.174$, $+0.046$, $+0.021$, and -0.039 across the four model/dataset pairs. Late hidden states are not automatically better routing representations.

The secondary relationship is therefore

$$S^* = f(\text{task, model, granularity, budget, stage})$$

within the measured scope. This is a reason to expose configuration dimensions, not evidence for a universally optimal setting. Detailed matched-token rows and the reachability audit appear in Appendix I.

8 Discussion

Meaning and identity are complementary. Semantic gists recover contextual relations, while exact, BM25, and approximate channels preserve different surface evidence. Their stable cross-model dataset ordering in Gate B makes heterogeneity more than a one-tokenizer curiosity.

Static fusion is not enough. Channels retrieve complementary evidence and complementary distractors. Under a small working-set budget, uncalibrated union or rank fusion can spend scarce

slots on the latter. The useful object is an explicit retrieval action with provenance, not a larger undifferentiated score vector.

Sparse discovery is feasible, but the fuzzy path is unfinished. Posting lists preserve the measured Top-4 outputs while reducing candidate work and exact-query latency sharply. Character and edit-based fallback still consumes hundreds of milliseconds in Python, and the current immutable rebuild is not an incremental production index.

Retrieval and consumption are distinct bottlenecks. HotpotQA supplies the positive bridge: correct sparse memory improves frozen-model likelihood and beats wrong memory. QASPER supplies the limiting case: high retrieval recall does not establish a fixed-channel gain over disabled memory. Later PRA work must study admission, attention competition, model adaptation, and training alongside retrieval.

Representation belongs in the control interface. Root and successor choices differ, and channel preference interacts with granularity and budget. The versioned action contract preserves independent root/successor methods, confidence status, provenance, and cost fields for adaptive control in Paper 3.5 [9]. Paper 3 separately controls how selected identities become physical K/V [10].

9 Related Work

Sparse, dense, and hybrid retrieval. BM25 and dense passage retrieval represent established lexical and semantic retrieval families [8, 3]. RAG supplies retrieved text to a generator, while reciprocal-rank fusion combines ranked lists without assuming comparable score scales [4, 2]. Hybrid IR is not our novelty. We study how these representation families address stable PRA memory identities under a native-K/V working-set budget, and we retain the negative result that static fusion need not beat its strongest component.

Multi-hop retrieval. MDR conditions later dense retrieval on earlier passages; IRRR builds a reasoning path, forms a next search query, reranks, and decides whether to continue; IRCOT interleaves retrieval with generated reasoning [17, 7, 14]. PRA similarly changes state after admission, but its search operates over identities aligned to reusable native-memory objects. Our narrower result is that the representation itself can change between root and successor stages.

Reference resolution. Approximate matching handles insertions, deletions, substitutions, and related surface changes [6]. Contextual entity linking, such as BLINK’s bi-encoder retrieval and cross-encoder reranking, addresses the harder problem of resolving ambiguous mentions [15]. Our token-native resolver is intentionally lighter. Its typo and alias gains coexist with stale-alias and wrong-instance failures, so it should not be read as a full entity linker.

Long-context and K/V memory. RETRO retrieves neighboring text chunks into a trained decoder, while Memorizing Transformers query a store of prior activation keys and values [1, 16]. Quest and RetrievalAttention perform query-aware physical K/V selection or indexing [13, 5]. Paper 2.6 operates primarily upstream, discovering identity-bearing logical objects, then uses Gate

B to verify their native-K/V consumption. The distinction is

$$\boxed{\text{logical memory discovery}} \neq \boxed{\text{physical K/V selection or compression}}.$$

10 Limitations

Gate B covers two small frozen model families and only HotpotQA/QASPER. Its evidence-first bounded construction is annotation dependent and does not reproduce an unmodified document benchmark. 2Wiki and MuSiQue appear only in the smaller legacy frozen-feature breadth study. The interaction cohort has eight examples per dataset. Five selector seeds measure optimization variability, not independent sample uncertainty.

Generated EM/F1 is weak, and the positive end-to-end claim is limited to paired gold-answer likelihood on HotpotQA. QASPER fixed-channel consumption remains unresolved. Natural top-reference correctness is only an evidence-overlap proxy, not adjudicated coreference. Validation-fitted confidence does not transfer reliably, confidently wrong references remain possible, and no learned selector closes the fixed-channel gap.

The index reports serialized payload bytes rather than Python-object memory. It rebuilds corpus statistics on update. Exact lookup is promising, but 0.50–0.65 s fuzzy routes are not production ready. The study does not test larger models, learned contextual entity linking, incremental serving indexes, larger selector-training sets, or PRA-aware fine-tuning. Failure to reproduce indexed parity, cross-model channel ordering, or the paired correct-versus-wrong-memory effects would falsify the corresponding claims independently.

11 Conclusion

One retrieval representation does not suffice for PRA memory discovery. Semantic meaning, exact identity, BM25-weighted terms, and approximate surface matching recover different evidence, and their dataset-level preferences replicate across the two tested model/tokenizer families. Sparse posting-list narrowing preserves measured Top-4 behavior while substantially reducing prototype work. Correctly discovered HotpotQA memory then produces a causal frozen-model likelihood benefit after native-K/V admission.

The boundary is equally important. QASPER shows that high evidence recall does not guarantee useful fixed-channel consumption. Root and successor search can prefer different representations, static fusion does not reliably solve that heterogeneity, and simple observable selectors remain inadequate. Paper 2.6 therefore solves neither adaptive control nor consumer training. It supplies a portable, indexed, multi-representational discovery interface and establishes where retrieval succeeds and where retrieval-to-consumption remains open.

A Original Discovery Experiment

The original Qwen frozen-feature experiment used eight validation and 16 held-out examples per dataset, layer-27 attention-input states, 32-token mean gists, and four requested chunks. It predates Gate B and is retained for provenance rather than treated as primary evidence. Broad semantic Top-8 deliberately used twice the request budget; the oracle admitted annotated evidence and is only a ceiling.

Table 6: Original held-out discovery. R/P/M are supporting-evidence recall, precision, and MRR. B0–H5 request four chunks; A1 requests eight.

Condition	HotpotQA			QASPER		
	R	P	M	R	P	M
gist	.151	.172	.276	.038	.062	.130
BM25	.398	.453	.875	.042	.094	.135
exact	.287	.328	.568	.178	.312	.443
weighted	.287	.328	.568	.061	.109	.214
approximate	.301	.359	.734	.037	.078	.141
union	.209	.250	.365	.038	.062	.130
token→semantic	.287	.344	.635	.044	.078	.125
semantic→token	.279	.328	.635	.052	.094	.156
cascade	.240	.281	.474	.126	.203	.307
iterative hybrid	.279	.312	.646	.081	.125	.177
broad semantic	.310	.180	.363	.088	.078	.155
oracle	.876	1.000	1.000	.633	1.000	1.000

Iterative hybrid improves HotpotQA gist recall by 0.129, with paired 95% bootstrap interval $[0.018, 0.242]$, but remains below BM25. Its QASPER improvement over gist is 0.043 with interval $[-0.014, 0.111]$; exact instead improves by 0.140 with interval $[0.053, 0.244]$. These small-cohort results anticipated representation heterogeneity but are superseded by Gate B for HotpotQA/QASPER.

B Full Four-Dataset Geometry

Best-channel paired intervals in the breadth cohort are QASPER $[.096, .271]$, HotpotQA $[.282, .516]$, 2Wiki $[.176, .353]$, and MuSiQue $[.122, .194]$. High IDF-weighted query/evidence overlap is associated with a 0.200 best-lexical-over-gist advantage, compared with 0.131 below the median. Examples with at least the median three distinct root choices have mean lexical advantage 0.197, versus 0.101 below the median. These are gold-derived explanatory strata, not deployable selector inputs.

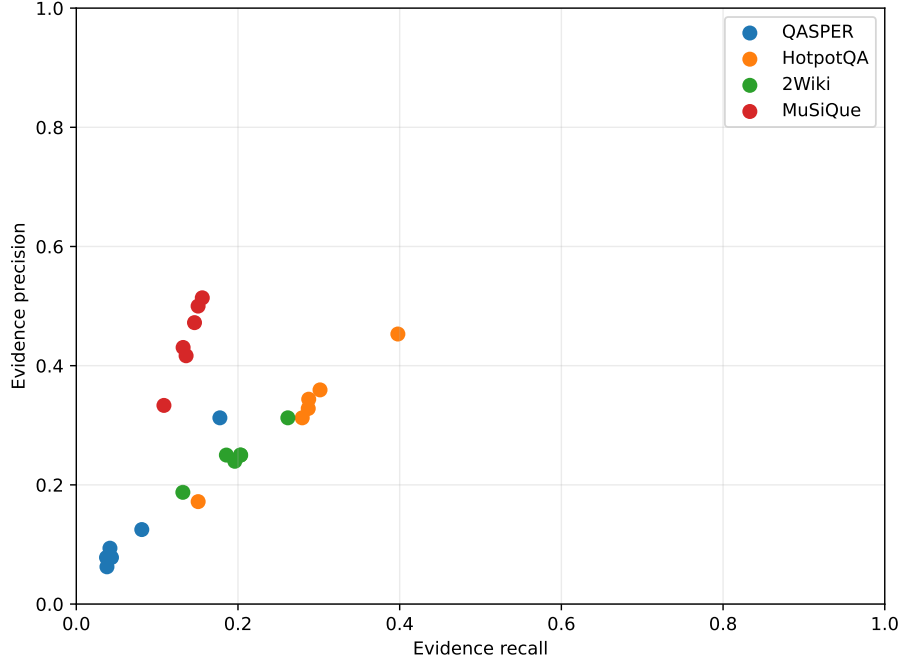


Figure 8: Full precision–recall surface for the four-dataset breadth cohort. Improved recall through union is not automatically a better policy when precision falls.

The per-example four-chunk oracle reaches recall 0.207, 0.489, 0.315, and 0.199 on QASPER, HotpotQA, 2Wiki, and MuSiQue. Earlier “headroom” values mixed true adaptive opportunity with validation-selected channel reversal; Appendix C gives the corrected decomposition. Full per-example rows, pairwise Jaccard matrices, lexical strata, and disagreement diagnostics are committed under `channel_geometry/`.

C Selectors, Confidence, and Referential Validity

At the two-chunk root stage, true per-example selection headroom $H_{selection}$ is 0.014, 0.021, 0.053, and 0.013 across QASPER, HotpotQA, 2Wiki, and MuSiQue. Validation instability $H_{validation}$ is 0, 0.031, 0.026, and 0. The corresponding successor values are 0.027/0, 0.028/0.056, 0.056/0.011, and 0.016/0.008. Only the 2Wiki root and MuSiQue successor paired intervals exclude zero. The action opportunity is real but small calibration splits cannot choose it reliably.

The leakage-audited replay exports 1,255 candidate rows with top score, score gap, entropy, support, span length, rarity, ambiguity, cross-channel agreement, semantic consistency, path consistency, and source consistency. Dataset identity, answer text, evidence geometry, and oracle labels are forbidden inputs. Validation-CDF confidence is poorly calibrated: mean root AUROC is 0.424, mean successor AUROC 0.526, and mean ECE 0.401 and 0.456. A semantic consistency gate never improves root top-one correctness over lexical confidence. The observable UsefulAddress proxy reaches AUROC 0.555 and AUPRC 0.230 over 68 successor-bearing identities.

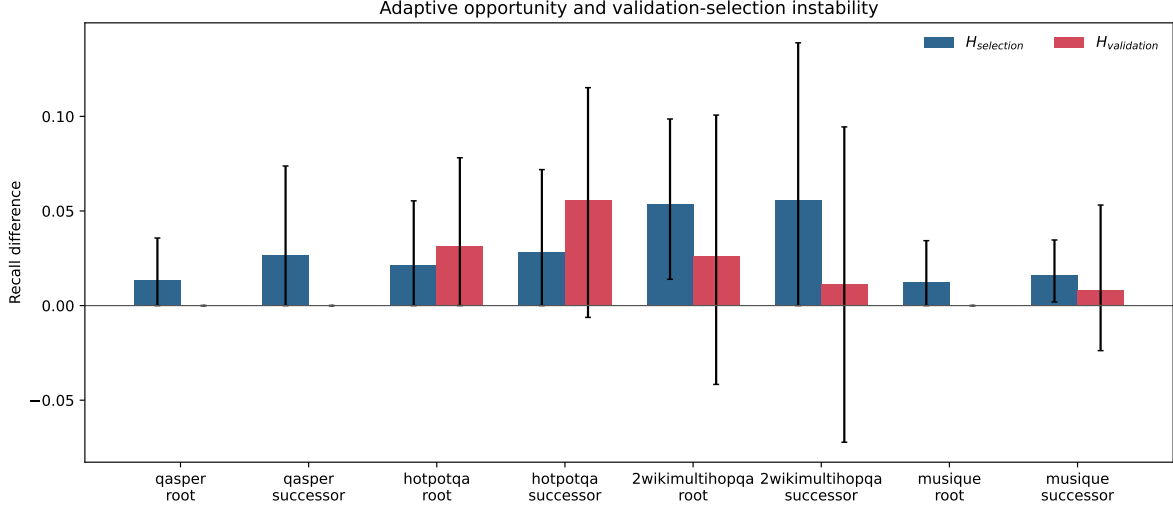


Figure 9: True selection opportunity and validation instability with identity-paired bootstrap intervals. Oracle effects remain diagnostic ceilings.

Natural Gate B top-reference accuracy further shows that retrieval score is not referential probability. Approximate top-one accuracy is 0.94 on HotpotQA but only 0.36–0.40 on QASPER. Except for approximate HotpotQA, validation thresholds targeting 0.80 precision transfer below 0.64 and often below 0.20 held-out precision.

D New Addresses and Controlled Chains

A strict new address is absent from the question, present in an admitted root and unrecovered gold evidence, and rare under local IDF. Such addresses appear in 6/16 QASPER, 5/16 HotpotQA, 10/24 2Wiki, and 15/18 MuSiQue examples. Exposure alone does not predict iterative gain. The stricter UsefulAddress predicate additionally requires a gold-linked successor to rank within four; it predicts positive hybrid-state gain only in the measured MuSiQue subset.

Twenty deterministic trials per condition test clean, case, punctuation, typo, alias, and conflicting-reference chains. Iterative hybrid recovers every clean, case, and punctuation target, half of typo targets, and none when the exposed address confidently names the wrong object. Exact search fails on typos; approximate search repairs every typo and half of alias cases, but follows stale aliases and wrong instances. These fixtures demonstrate mechanism and failure modes, not natural-frequency estimates.

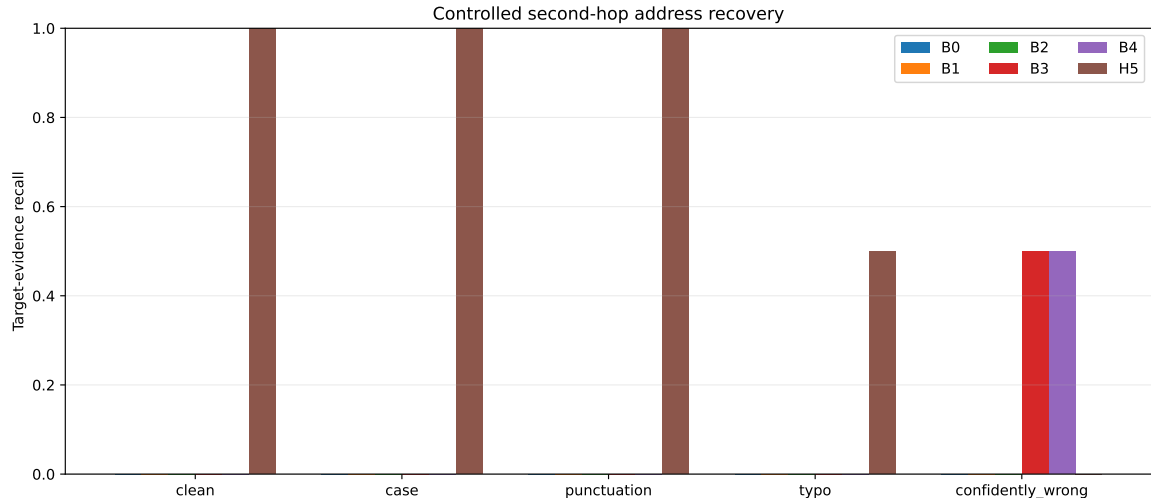


Figure 10: Controlled second-hop recovery. A newly exposed address can enable retrieval, but confidently wrong references produce catastrophic failure.

E Tokenizer and Surface-Form Robustness

The expanded suite covers case, punctuation, leading-space artifacts, whitespace splitting, hyphenation, concatenation, misspellings, abbreviations, aliases, Unicode compatibility forms, near entities, shared prefixes, identifiers, URLs, same-name and same-class ambiguity, wrong relations, stale aliases, and contradictory high-overlap names. Across 13 primary perturbations, cascade recall@4 is 0.985 with corpus IDF and 0.969 with no suppression or a fixed stop list, identically for Qwen and SmolLM tokenizers. Exact reaches 0.923, token n -grams 0.754, approximate 0.262, and bounded edit 0.246.

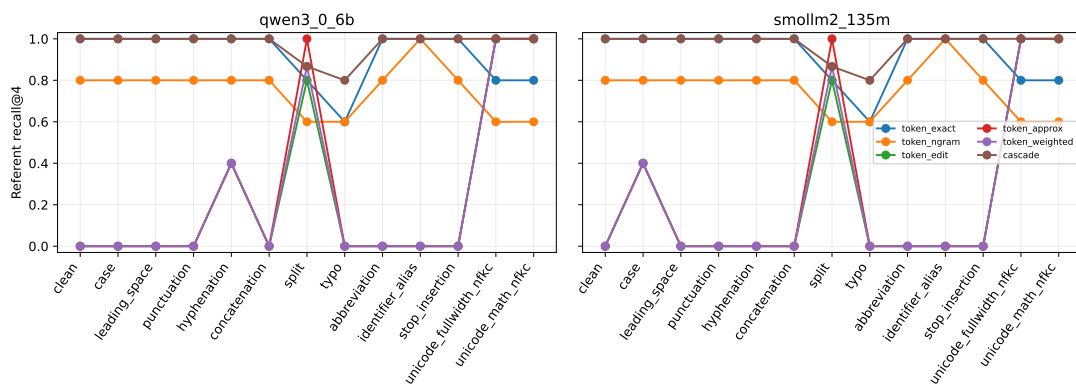


Figure 11: Controlled target recovery across tokenizers and perturbations. NFKC handles compatibility forms; fallback stages cover some genuinely changed sequences.

F Wrong-Reference Fixtures

The controlled suite includes stale aliases, wrong instances, wrong relations, same-name collisions, contradictory high-overlap references, and two plausible references. Stale aliases route to the wrong

target in every exact, approximate, and hybrid trial. Approximate search selects the same-class wrong instance in every trial; exact and hybrid do so in half. Under a correct-entity/wrong-relation fixture, approximate search splits between valid and wrong targets while exact and hybrid recover neither. Threshold rejection creates a fixture-specific abstention opportunity, but no threshold is deployment calibrated.

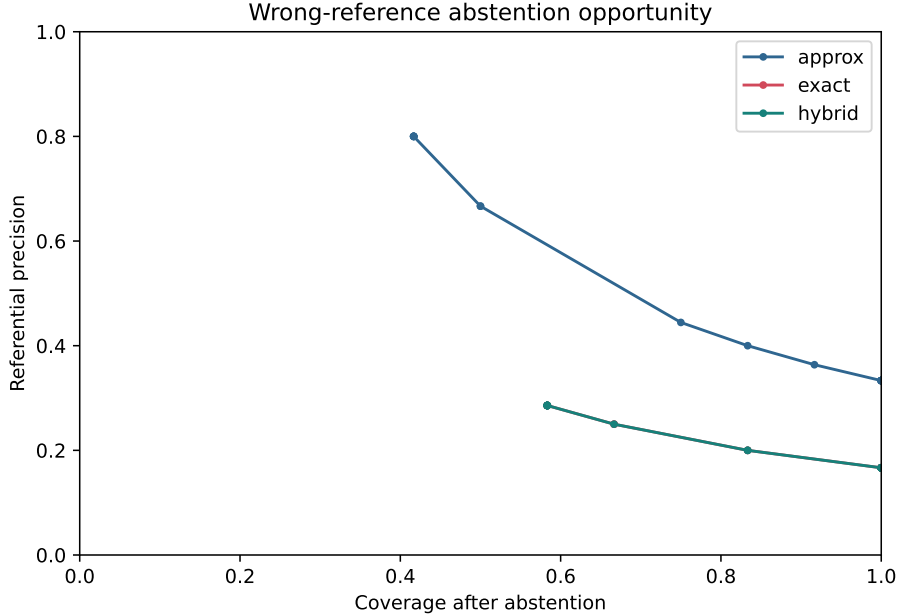


Figure 12: Controlled referential precision against retained coverage. This is an abstention opportunity bound, not a natural policy result.

G Indexed Implementation and Cost

Table 7: Warm routing at 4,096 chunks. Milliseconds are per query; payload MiB excludes Python-object overhead.

Tokenizer	Channel	Exhaustive	Indexed	Speedup	Payload MiB
Qwen	exact	33557	119	283x	10.06
Qwen	weighted	31375	530	59x	10.06
Qwen	approximate	31141	499	62x	10.06
Qwen	cascade	30965	556	56x	10.06
SmolLM	exact	37048	125	300x	10.94
SmolLM	weighted	37440	649	58x	10.94
SmolLM	approximate	37005	632	59x	10.94
SmolLM	cascade	36886	577	64x	10.94

`TokenNativeIndex` in `src/prahf/hybrid_discovery.py` owns aligned rows, tokenization, postings, aliases, payload estimates, immutable rebuilds, and channel scoring. Its candidate-union method reserves semantic rows, inserts explicit URI rows, and deduplicates stable identities. Expensive feature families are gated by the requested channel. `IterativeGistRouter.route` in `src/prahf/iterative.py` performs root and successor sparse scoring, scatters returned rows into

the aligned tensor index, and reuses the inherited visited-set, branch, beam, depth, confidence, and unique-chunk limits.

PRAForCausalLM constructs the sidecar, validates explicit reference URIs, runs discovery, maps selected conceptual identities across consumption layers, and only then marks K/V as materialized. The public generation API records resolved URI state and physical statistics. Explicit URI priority is therefore a deterministic caller-supplied path, not evidence that the router inferred the correct reference.

H Full Root–Successor Surface

Every one of five root actions is crossed with five successor actions. Per-example oracle transitions are heterogeneous: the most frequent are exact→exact (9), exact→native semantic (8), exact→BM25-state (7), gist→native semantic (6), and BM25→native semantic (5). No transition is a majority. The corresponding path gains are 0.038, 0.105, 0.076, 0.065, and 0.062.

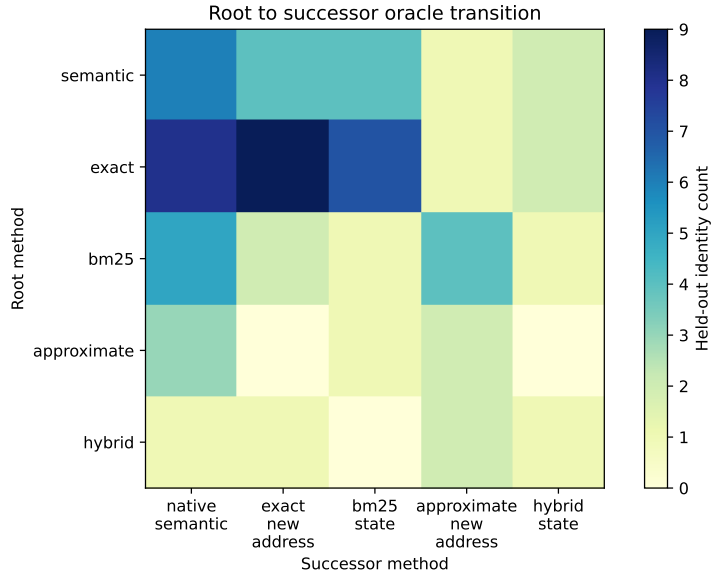


Figure 13: Per-example oracle root-to-successor transitions in the breadth cohort. The matrix is an opportunity diagnostic, not a deployable controller.

Table 8: Normalized efficiency for the highest-recall measured root–successor pair in each breadth cohort.

Dataset	Root → successor	R_E	P_E	K/N	K/E
QASPER	exact → exact address	.121	.219	.032	1.029
HotpotQA	approximate → semantic	.145	.156	.088	.896
2Wiki	exact → semantic	.148	.236	.102	.901
MuSiQue	BM25 → exact address	.086	.278	.050	.327

Complete recovery is zero throughout. Search comparisons and physical K/V activation remain separate costs.

I Interaction Details

The corrected interaction run contains 4,608 rows and makes Top-2, Top-4, and Top-8 reachable by setting per-hop branch and beam widths to 1, 2, and 4. A pilot silently capped Top-8 and is excluded. Table 9 reports exploratory maxima at a fixed 128 routed tokens.

Table 9: Best late-layer channel at 128 routed tokens. C_E is complete evidence recovery.

Model	Dataset	Chunk	Best channel	Recall	Precision	C_E
Qwen	HotpotQA	16	token n -gram	.938	.266	.875
Qwen	HotpotQA	32	bounded edit	.750	.281	.625
Qwen	HotpotQA	64	token n -gram	.875	.438	.875
Qwen	QASPER	16	token n -gram	1.000	.547	1.000
Qwen	QASPER	32	bounded edit	1.000	.812	1.000
Qwen	QASPER	64	exact	1.000	.938	1.000
SmolLM	HotpotQA	16	bounded edit	.938	.328	.875
SmolLM	HotpotQA	32	exact	.812	.312	.625
SmolLM	HotpotQA	64	token n -gram	1.000	.500	1.000
SmolLM	QASPER	16	hybrid	.958	.438	.875
SmolLM	QASPER	32	token n -gram	.958	.625	.875
SmolLM	QASPER	64	exact	1.000	.875	1.000

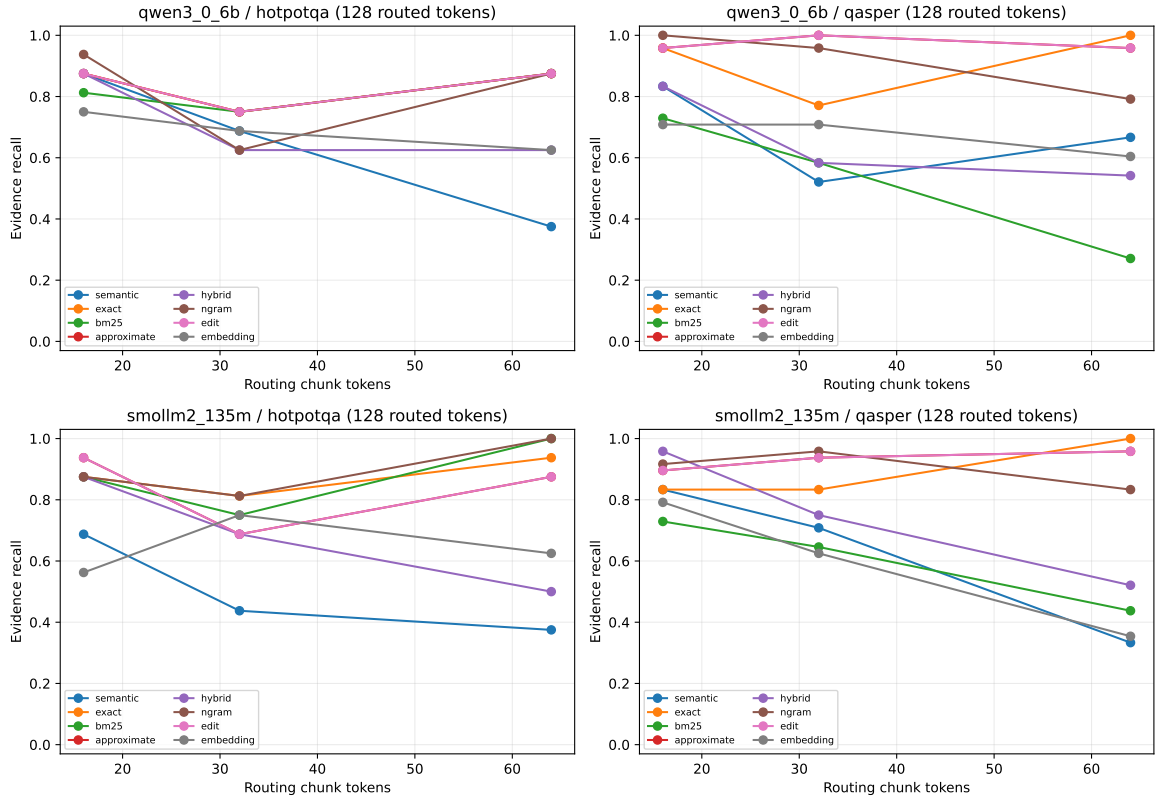


Figure 14: Evidence recall by representation and chunk size at 128 routed tokens. Ordering changes across dataset, model, and granularity.

J Artifact and Action Contract

The version-2 action specification freezes five root actions—semantic, exact, BM25, approximate, and hybrid—and five successor actions—native semantic, exact new address, BM25 state, approximate new address, and hybrid state. Each action names its implementation identifier, required state and indexes, legal parameters, confidence outputs, cost metrics, and known failures. Legacy names remain explicit aliases. The schema records whether a source experiment materialized K/V so that discovery-only artifacts cannot be mistaken for consumption measurements.

Committed artifacts include per-example retrieval rows, cohort-overlap audits, controller seed rows, paired causal effects, physical K/V accounting, indexed latency and parity summaries, tokenizer perturbations, complete root–successor matrices, interaction rows, confidence diagnostics, and claim audits. The canonical locations are `docs/papers/shared/results/paper2_6_hybrid_pra/` and `experiments/paper2_6_hybrid_pra/`. This contract lets Paper 3.5 consume actions and observable fields without importing dataset identity or gold evidence into its controller.

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